

Dart Learning Roadmap for a JavaScript Developer

Duration: 8 Weeks

Prerequisite: JavaScript (ES6+)

Outcome: Build Dart applications and become productive with Flutter.

Week 1: Dart Fundamentals

Goals

- Understand Dart syntax
- Learn variables, types, and null safety
- Work with functions and collections

Resources

Official Documentation

- [Dart Language Tour](#)

Video Course

- [Dart Programming Tutorial for Beginners \(freeCodeCamp\)](#)

Study Schedule

Day 1

- 00:00–30:00 Variables & Data Types
- 30:00–55:00 Null Safety

Day 2

- 55:00–01:30:00 Functions
- 01:30:00–02:00:00 Control Flow

Day 3

- 02:00:00–02:45:00 Collections (List, Set, Map)
- 02:45:00–End Review

Project

Build a simple:

- Expense Tracker
 - Student Grade Calculator
-

Week 2: Object-Oriented Dart

Goals

- Learn Dart's OOP model
- Master classes and inheritance

Topics

- Classes
- Constructors
- Inheritance
- Abstract Classes
- Mixins
- Extensions

Resource

- [Dart Classes Guide](#)

Project

Library Management System

Features:

- Add Books
 - Remove Books
 - Search Books
-

Week 3: Async Programming

Goals

- Understand Futures and Streams
- Use `async/await` confidently

Resource

- [Dart Async Guide](#)

Schedule

Day 1

- Futures

Day 2

- `async/await`

Day 3

- Streams

Day 4

- Error Handling

Project

Crypto Price Tracker

Skills:

- API Requests
- Async Operations
- Error Handling

Week 4: Advanced Dart

Goals

Learn modern Dart language features.

Topics

- Generics
- Records
- Pattern Matching
- Enums
- Sealed Classes

Resource

- Dart Language Evolution

Project

Task Manager CLI

Requirements:

- Classes
 - Async Functions
 - Pattern Matching
 - Generics
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Week 5: Flutter Basics

Resource

- [Flutter Get Started Guide](#)

Video

- [Flutter Course for Beginners \(freeCodeCamp\)](#)

Timestamps

00:00–02:00

- Flutter Installation
- Widgets
- Layouts

Project

Simple Todo App

Week 6: State Management & Navigation

Timestamps

02:00–04:30

- State Management Basics
- Navigation
- Routing

Project

Notes App

Week 7: APIs & Forms

Timestamps

04:30–07:00

- API Integration
- Forms
- Validation
- User Input

Project

Weather App

Week 8: Storage & Deployment

Timestamps

07:00–End

- Local Storage
- Shared Preferences
- Deployment

Project

Expense Tracker App

Portfolio Projects

Beginner

1. Calculator
2. Todo App
3. Weather App

Intermediate

1. Expense Tracker
2. Notes App
3. Movie Search App

Advanced

1. Chat Application
 2. E-commerce App
 3. Fitness Tracker
 4. Ride-Hailing Clone
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Daily Study Routine (90 Minutes)

15 Minutes

Read official Dart/Flutter documentation

45 Minutes

Watch tutorials

30 Minutes

Build something from memory without copying

Fast-Track Path (Recommended)

Week 1

- Dart Language Tour
- Basic Syntax

Week 2

- OOP
- Async Programming

Week 3

- CLI Projects

Weeks 4–8

- Flutter
- Portfolio Projects

Expected Result: Productive in Dart within 3 weeks and capable of building Flutter applications within 2 months.